🍓 AGRIBOT - Raspberry Pi Zero 2 W Complete Setup

📋 Part 1: Fresh Raspberry Pi OS Installation

Step 1: Flash Raspberry Pi OS

1. **Download Raspberry Pi Imager:**
   - https://www.raspberrypi.com/software/
2. **Flash OS:**
   - Open Raspberry Pi Imager
   - Choose Device: **Raspberry Pi Zero 2 W**
   - Choose OS: **Raspberry Pi OS Lite (64-bit)** (recommended for headless)
   - Choose Storage: Your SD Card
   - Click **NEXT**
3. **Important Settings (Click "Edit Settings"):**

```
✅ Set hostname: agribot
✅ Enable SSH: YES
✅ Set username: pi
✅ Set password: [your-password]
✅ Configure WiFi:
   - SSID: [your-wifi-name]
   - Password: [your-wifi-password]
✅ Set timezone: Asia/Dhaka
```

4. **Write to SD Card**
   - Click **YES** to apply settings
   - Click **YES** to erase
   - Wait for completion
5. **Insert SD card into Raspberry Pi and boot**

📋 Part 2: First Boot & SSH Access

Step 1: Find Raspberry Pi IP Address

Method 1 - Using Router:

```
- Login to your router (192.168.1.1 or 192.168.0.1)
- Look for device named "agribot"
- Note the IP address
```

Method 2 - Using Phone App:

```
- Install "Fing" app on phone
- Scan network
- Find "Raspberry Pi" device
```

Method 3 - Using Computer:

bash

```
# Linux/Mac
ping agribot.local

# Windows
ping agribot.local
```

Step 2: SSH into Raspberry Pi

bash

```
# From your computer
ssh pi@agribot.local
# OR
ssh pi@192.168.1.XXX

# Password: [যা আপনি set করেছিলেন]
```

📋 Part 3: Initial Configuration

Step 1: Update System

bash

```
# Update package lists
sudo apt update

# Upgrade all packages
sudo apt upgrade -y

# This may take 10-15 minutes
```

Step 2: Enable Hardware Interfaces

bash

```
# Open configuration
sudo raspi-config

# Navigate using arrow keys:

# 1. Interface Options → Serial Port
#    - Login shell over serial: NO
#    - Serial port hardware: YES

# 2. Interface Options → I2C
#    - Enable: YES

# 3. Performance Options → GPU Memory
#    - Set to: 128

# 4. Localisation → Timezone
#    - Asia → Dhaka

# Select <Finish>
# Reboot: YES
```

Step 3: Configure Boot Settings

bash

```
# Edit boot config
sudo nano /boot/config.txt

# Add these lines at the end:
enable_uart=1
dtoverlay=disable-bt
gpu_mem=128
start_x=1

# Save: Ctrl+X, then Y, then Enter

# Edit cmdline (for serial)
sudo nano /boot/cmdline.txt

# Remove these if present:
# console=serial0,115200
# (Just delete that part)

# Save and exit
```

Step 4: Reboot

bash

```
sudo reboot
```

Wait 1 minute, then SSH again.

📋 Part 4: Install Required Packages

Step 1: Install System Packages

bash

```
# Install all at once
sudo apt install -y \
    python3-pip \
    python3-dev \
    python3-opencv \
    git \
    nginx \
    vim \
    nano \
    screen \
    build-essential \
    cmake \
    v4l-utils

# This takes 5-10 minutes
```

Step 2: Install Python Packages

bash

```
# Upgrade pip first
pip3 install --upgrade pip

# Install required packages
pip3 install \
    websockets \
    pyserial \
    requests \
    numpy \
    opencv-python-headless

# This takes 5-10 minutes on Pi Zero 2 W
```

Step 3: Add User to Groups

bash

```bash
# Add pi user to required groups
sudo usermod -a -G dialout pi
sudo usermod -a -G video pi
sudo usermod -a -G gpio pi

# Verify
groups pi

# Should show: pi adm dialout video gpio sudo ...
```

📋 Part 5: Create Project Structure

bash

```bash
# Create main directory
mkdir -p ~/agribot_data/evidence
mkdir -p ~/agribot_data/recordings

# Verify
ls -la ~/

# You should see:
# drwxr-xr-x agribot_data
```

📋 Part 6: Setup Serial Ports

Check Available Ports

bash

```bash
# List all serial devices
ls -l /dev/tty*

# You should see:
# /dev/ttyAMA0  (GPIO UART for GPS)
# /dev/ttyACM0  (Arduino - when connected)
# /dev/ttyUSB0  (GSM - when connected)
```

Test Serial Access

```bash
# Give permission (temporary)
sudo chmod 666 /dev/ttyAMA0

# Test GPS (if connected)
cat /dev/ttyAMA0
# Should show NMEA sentences like $GPGGA...
# Press Ctrl+C to stop
```

📋 Part 7: Upload Agribot Files

Method 1: Using SCP (from your computer)

```bash
# From your computer, not from Pi!

# Copy Python main code
scp agribot_main.py pi@agribot.local:~/

# Copy HTML file
scp agribot_control.html pi@agribot.local:~/

# If you have multiple files:
scp agribot_*.py pi@agribot.local:~/
```

Method 2: Using Nano (create files directly on Pi)

```bash
```

```
# SSH into Pi
ssh pi@agribot.local

# Create main Python file
nano ~/agribot_main.py

# Paste the complete agribot_main.py code here
# Save: Ctrl+X, Y, Enter

# Create HTML file
nano ~/agribot_control.html

# Paste the HTML code
# Save: Ctrl+X, Y, Enter

# Make executable
chmod +x ~/agribot_main.py
```

📋 Part 8: Setup Web Server

bash

```
# Copy HTML to nginx directory
sudo cp ~/agribot_control.html /var/www/html/index.html

# Set permissions
sudo chown www-data:www-data /var/www/html/index.html
sudo chmod 644 /var/www/html/index.html

# Start nginx
sudo systemctl enable nginx
sudo systemctl start nginx

# Check status
sudo systemctl status nginx

# Should show: active (running)
```

Test Web Interface

```bash
# Get Pi IP address
hostname -I

# Open browser on your phone/computer:
http://192.168.1.XXX

# You should see the Agribot control interface
```

📋 Part 9: Create Systemd Service (Auto-start)

Create Service File

```bash
sudo nano /etc/systemd/system/agribot.service
```

Paste this:

```ini
[Unit]
Description=Agribot Ultimate Control System
After=network.target

[Service]
Type=simple
User=pi
WorkingDirectory=/home/pi
ExecStart=/usr/bin/python3 /home/pi/agribot_main.py
Restart=always
RestartSec=10
StandardOutput=append:/home/pi/agribot_data/agribot.log
StandardError=append:/home/pi/agribot_data/agribot_error.log

[Install]
WantedBy=multi-user.target
```

Save: Ctrl+X, Y, Enter

Enable Service

```bash
# Reload systemd
sudo systemctl daemon-reload

# Enable service (auto-start on boot)
sudo systemctl enable agribot.service

# DON'T start yet - we need to connect hardware first
```

📋 Part 10: Connect Hardware

Power Off Pi

```bash
sudo poweroff
```

Wait for green LED to stop blinking, then disconnect power.

Connect Hardware (Pi Powered OFF!)

GPS NEO-6M:

```
GPS TX  → Pi Pin 10 (GPIO15 RXD)
GPS RX  → Pi Pin 8  (GPIO14 TXD)
GPS VCC → Pi Pin 2  (5V)
GPS GND → Pi Pin 6  (GND)
```

Arduino UNO:

```
Connect via USB cable to Pi USB port
```

GSM SIM900A:

```
GSM Module → TTL-to-USB Adapter → Pi USB port
(Make sure GSM has 12V power from battery!)
```

USB Camera:

```
Connect to Pi USB port
```

Power:

```
Buck Converter 5V OUT → Pi Pin 2 or 4 (5V)
Buck Converter GND    → Pi Pin 6 (GND)
```

📋 Part 11: First Boot with Hardware

Power On

bash

```bash
# Connect 5V power to Pi
# Wait 30 seconds for boot
# Green LED will blink during boot

# SSH into Pi
ssh pi@agribot.local
```

Verify Hardware Connections

bash

```
# Check Arduino
ls -l /dev/ttyACM0
# Should exist

# Check GPS
ls -l /dev/ttyAMA0
# Should exist

# Check GSM
ls -l /dev/ttyUSB0
# Should exist

# Check Camera
ls -l /dev/video0
# Should exist
```

Test Each Component

Test GPS:

bash

```
cat /dev/ttyAMA0
# Should show: $GPGGA,$GPRMC sentences
# Ctrl+C to stop
```

Test GSM:

bash

```
screen /dev/ttyUSB0 9600
# Type: AT
# Press Ctrl+J
# Should respond: OK
# Exit: Ctrl+A, then D
```

Test Arduino:

bash

```
screen /dev/ttyACM0 9600
# Should show: AGRIBOT_ULTIMATE_READY
# Type: STATUS
# Should show sensor data
# Exit: Ctrl+A, then D
```

Test Camera:

bash

```
v4l2-ctl --list-devices
# Should show camera

# Test capture
python3 << 'EOF'
import cv2
cap = cv2.VideoCapture(0)
if cap.isOpened():
    print("✓ Camera working")
    cap.release()
else:
    print("✗ Camera failed")
EOF
```

📋 Part 12: Start Agribot System

Manual Start (for testing)

bash

```
# Start main system
python3 ~/agribot_main.py

# You should see:
# ✓ Configuration loaded
# ✓ Arduino connected
# ✓ GSM connected
# ✓ GPS connected
# ✓ Camera connected
# 🚀 Starting all systems...

# Keep this running in one terminal
```

Open Another Terminal (for logs)

bash

```
# New SSH session
ssh pi@agribot.local

# Watch activity log
tail -f ~/agribot_data/activity_log.txt
```

Test Web Control

bash

```
# On your phone/computer browser:
http://192.168.1.XXX

# Try:
- Click FORWARD button
- Robot should move forward
- Click STOP
```

📋 Part 13: Enable Auto-Start

If everything works in manual mode:

```bash
# Stop manual mode (Ctrl+C)

# Start as service
sudo systemctl start agribot.service

# Check status
sudo systemctl status agribot.service

# Should show: active (running)

# View logs
sudo journalctl -u agribot.service -f
```

Reboot Test

```bash
# Reboot Pi
sudo reboot

# Wait 2 minutes

# Check if service started automatically
sudo systemctl status agribot.service

# Should be: active (running)
```

📋 Part 14: WiFi Management Setup

Create WiFi Change Script

```bash
nano ~/change_wifi.sh
```

Paste:

```bash
```

```
#!/bin/bash
echo "Current WiFi networks:"
sudo iwlist wlan0 scan | grep ESSID

echo ""
read -p "Enter WiFi SSID: " SSID
read -sp "Enter WiFi Password: " PASS
echo ""

sudo tee /etc/wpa_supplicant/wpa_supplicant.conf > /dev/null <<EOF
country=BD
ctrl_interface=DIR=/var/run/wpa_supplicant GROUP=netdev
update_config=1

network={
    ssid="$SSID"
    psk="$PASS"
    key_mgmt=WPA-PSK
}
EOF

echo "Restarting WiFi..."
sudo wpa_cli -i wlan0 reconfigure
echo "Done! New WiFi configured."
```

Make executable:

bash

```
chmod +x ~/change_wifi.sh

# Usage:
./change_wifi.sh
```

📋 Part 15: Backup & Recovery

Create Full Backup

bash

```bash
# Backup all config files
tar -czf ~/agribot_backup.tar.gz \
    ~/agribot_main.py \
    ~/agribot_control.html \
    ~/agribot_data/charge_location.json \
    /etc/systemd/system/agribot.service \
    /var/www/html/index.html

# Download to your computer
# From your computer:
scp pi@agribot.local:~/agribot_backup.tar.gz ./
```

Restore from Backup

bash

```bash
# Upload backup to Pi
scp agribot_backup.tar.gz pi@agribot.local:~/

# Extract
cd ~
tar -xzf agribot_backup.tar.gz

# Restore service
sudo cp ~/agribot.service /etc/systemd/system/
sudo systemctl daemon-reload
sudo systemctl enable agribot.service
```

📋 Part 16: Troubleshooting Commands

bash

```bash
# Check system logs
sudo journalctl -xe

# Check service status
sudo systemctl status agribot.service

# Restart service
sudo systemctl restart agribot.service

# View live logs
tail -f ~/agribot_data/activity_log.txt

# Check disk space
df -h

# Check memory
free -h

# Check processes
htop

# Check temperature
vcgencmd measure_temp

# Restart networking
sudo systemctl restart networking

# Check WiFi
iwconfig wlan0
```

📋 Part 17: Performance Optimization

bash

```bash
# Disable unnecessary services
sudo systemctl disable bluetooth
sudo systemctl disable avahi-daemon

# Increase swap for better performance
sudo dphys-swapfile swapoff
sudo nano /etc/dphys-swapfile
# Change: CONF_SWAPSIZE=100 to CONF_SWAPSIZE=512
sudo dphys-swapfile setup
sudo dphys-swapfile swapon

# Clean old logs
sudo journalctl --vacuum-time=7d
```

✅ Final Checklist

bash

```bash
# Run this verification script:
python3 << 'VERIFY'
import os
import subprocess

checks = {
    "agribot_main.py exists": os.path.exists("/home/pi/agribot_main.py"),
    "HTML exists": os.path.exists("/var/www/html/index.html"),
    "Data directory exists": os.path.exists("/home/pi/agribot_data"),
    "Service enabled": "enabled" in subprocess.getoutput("systemctl is-enabled
}

print("\n=== Agribot Setup Verification ===\n")
for check, status in checks.items():
    symbol = "✓" if status else "✗"
    print(f"{symbol} {check}")

print("\n" + "="*35 + "\n")
VERIFY
```

🎉 Your Raspberry Pi is Ready!

Quick Start Commands:

```bash
# Start system
sudo systemctl start agribot.service

# Stop system
sudo systemctl stop agribot.service

# Restart system
sudo systemctl restart agribot.service

# View logs
tail -f ~/agribot_data/activity_log.txt

# Check status
sudo systemctl status agribot.service

# Change WiFi
./change_wifi.sh
```

Access Points:

```
Web Interface: http://RASPBERRY_PI_IP
SSH Access: ssh pi@agribot.local
Logs: ~/agribot_data/activity_log.txt
Recordings: ~/agribot_data/recordings/
```

এখন আপনার Raspberry Pi সম্পূর্ণ ready! 🚀

কোনো সমস্যা হলে বলুন! 😊